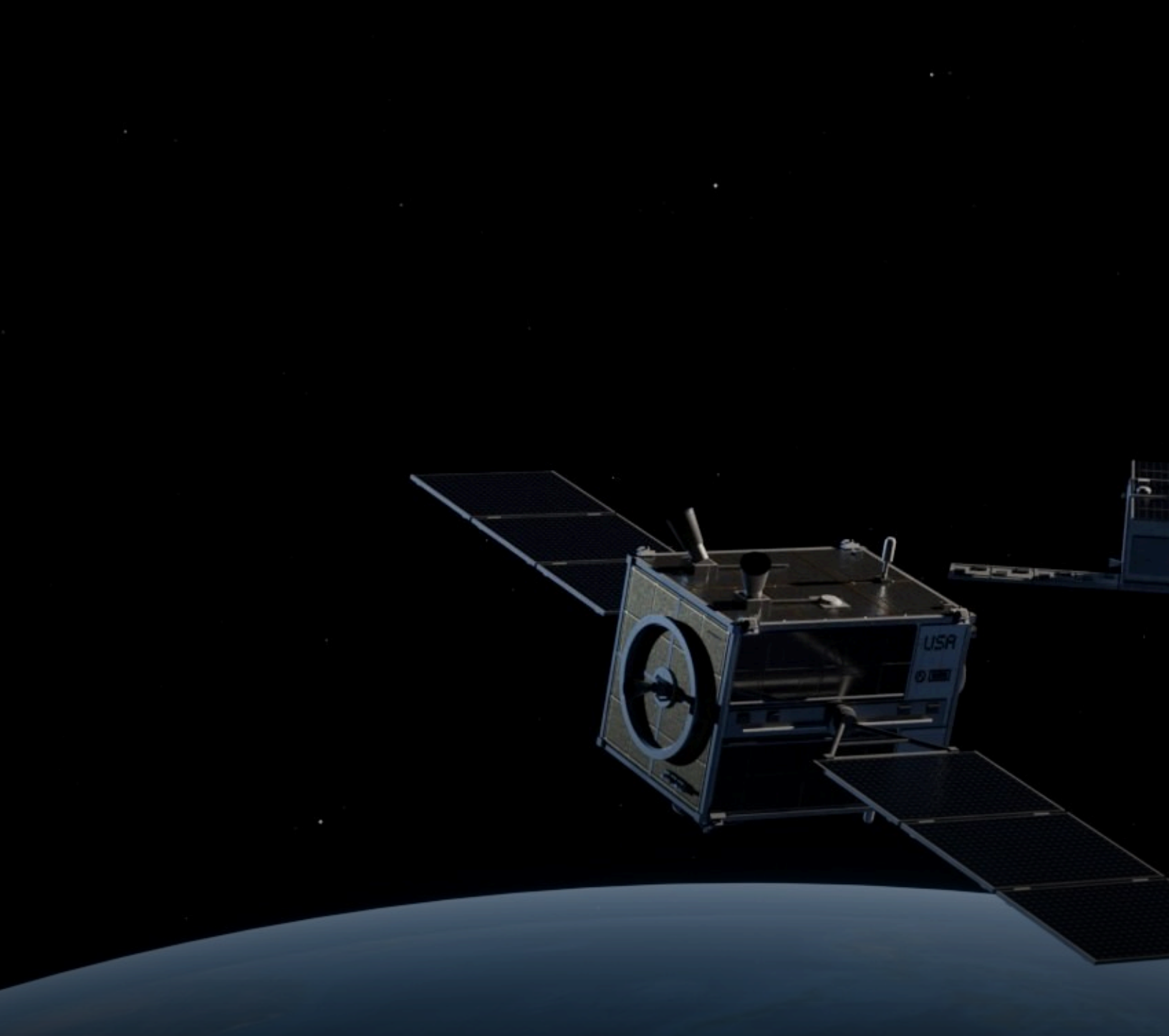


OPERATOR'S HANDBOOK

xGEO LIVE

The complete guide to xGEO Live: every screen, every station and every key, how the physics shapes your choices, and how to win CLOSE PASS in low Earth orbit.

ALPHA 0.5 · BUILD 11 · KEYBOARD & MOUSE · 38 PAGES

WELCOME

BEFORE YOU PLAY

xGEO Live is a game about the quiet contest between satellites. You fly one inspector satellite, watch another nation's inspector, and use real orbits, real sensors and real radio to make it leave without starting a war.

WHAT KIND OF GAME THIS IS

There are no lasers and no explosions. Satellites in low orbit move at nearly 8 km/s, but they meet and part slowly, a few metres a second at a time. Every move costs fuel you can never get back, every look depends on where the Sun is, and every radio burst you hear is worked out from real signal physics.

The game runs on **xGEO Core**, a simulation engine that computes orbits with the Earth's flattened shape and the pull of the Sun and the Moon. Nothing you see is scripted to look right: it is computed. When the handbook says something "costs 2.3 m/s", that number came out of the same physics a mission planner would use.

You don't need to know orbital mechanics. The planner does the maths and tells you the price. This handbook explains the few rules of thumb that make you better at it.

HOW TO USE THIS HANDBOOK

New to the game? Start the game and follow the tutorial card. It teaches the whole program in 24 short steps. Then read chapters 2 and 12 to learn the mission and your opponent.

Looking something up? Chapters 3 to 11 describe each screen and station in turn. The cheat sheet (page 36) and the glossary (page 37) are made for printing.

WHAT YOU NEED



A PC

Desktop or laptop. Phones and touch screens are not supported.



A BROWSER

Current Chrome, Edge or Firefox with WebGL 2.



MOUSE AND KEYBOARD

A wheel or trackpad for zooming. A middle button helps.



ABOUT 130 MB

The first load takes a minute. Your browser then keeps it.



1920 × 1080 OR LARGER

Smaller works, but windows overlap. Press F11 for full screen.

FIRST CLICK Click inside the game once after it loads, so it gets your keyboard. Some browsers keep F1 to F7 for themselves until you do.

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WORDS IN THIS HANDBOOK

Labels you see on screen are written `LIKE THIS`. Keys look like <kbd>F3</kbd>. The game can use plain words or military ones (see Terms, page 10). This handbook uses the plain words and gives the military one in brackets the first time.

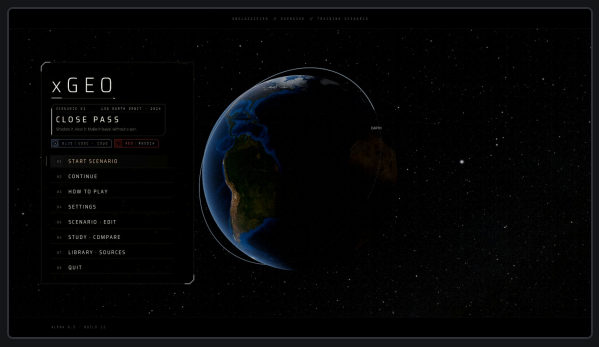
REAL AND INVENTED

ICEYE-X36 and Cosmos 2612 are real satellites, flown from their public orbits and published specifications. The five ground radars are real sites. WATCHMAN-1 is invented, loosely modelled on the US GSSAP inspectors, and what happens in the game is fiction.

1 · GETTING STARTED

THE TITLE SCREEN

The game opens on the live Earth turning slowly behind the menu. Any key or click skips the short intro.



The title screen. The block at the top names the scenario: SCENARIO 01, CLOSE PASS, low Earth orbit, 2026.

ITEM	WHAT IT DOES
START SCENARIO	A new game of CLOSE PASS, with the tutorial on. Time starts paused.
CONTINUE	Resumes your save. Greyed out when there is none.
HOW TO PLAY	A one-page briefing: the goals, the four things you do, and the keys.
SETTINGS	Terms (plain or doctrinal), music and effects volume.
SCENARIO · EDIT	Change the scenario's numbers, then play them (page 5).
STUDY · COMPARE	Run many games of each plan and compare them (page 5).
LIBRARY · SOURCES	Where the numbers come from, with why each source was chosen.
QUIT	Leaves the game.

PAGES BESIDE THE MENU

Each item except START, CONTINUE and QUIT opens a page beside the menu (under it, on a tall screen). Close a page with `ESC CLOSE`, the **Esc** key, or by clicking the same item again.

SAVING AND CONTINUING

There is one save. The game writes it:

- whenever you go back to the title screen (**Esc** with no window open, or `File › Save and go to the menu`);
- automatically, once for every day of mission time;
- with `File › Save and quit` (desktop program only).

The save is deleted when you leave the debrief, so a finished operation can't be continued. A continued game has no tutorial.

SETTINGS

SETTING	CHOICES
TERMS	**PLAIN** or **DOCTRINAL**. Plain words, or the terms of US joint and Space Force doctrine. The game is the same either way. You can also switch in game from the View menu.
MUSIC	A volume slider.
EFFECTS	Volume of clicks, alarms and drives.

> **NOTE** In this build the volumes reset each time the game starts, and Terms is kept only inside a save. Map layers, contact symbols and window sizes are remembered.

EDIT, STUDY, LIBRARY

xGEO is a simulation first. Two pages on the title screen let you change the scenario's numbers and test plans against each other over many runs.

SCENARIO · EDIT

Close Pass's main numbers, each with the scenario's own value shown beside it in grey. Change any of them and press `PLAY EDITED`. Edits are kept until you press `RESET`.

GROUP	WHAT YOU CAN CHANGE
START	How far behind X36 WATCHMAN-1 and Cosmos 2612 start (20 km and 5.1 km), the catalogue's error on Cosmos (0.2 km), the operation's length (2 days).
WATCHMAN-1	Propellant (75 kg), exhaust speed (2.2 km/s), telescope size (0.3 m).
COSMOS 2612'S BEHAVIOUR	When it gives way (5 km), how far it moves away (8 km), how many times it gives way before leaving (2), when it feels unwatched (50 km), the closest it will come to X36 (0.5 km).
RULES AND TENSION	X36's keep-out (1 km), what counts as a good track (1 km, 80 % of the time), tension per crowding (5) and per hour too close (12), when it leaves (50) and when war breaks out (100).
SCORE WEIGHTS	The points for each goal (40, 20, 20, 10, 10).

`SAVE AS FILE` writes the whole edited scenario to a file you can run with xGEO Core's command-line tools.

> **CAREFUL** START SCENARIO on the menu always plays the scenario as published. Only PLAY EDITED uses your edits.

STUDY · COMPARE

Plays CLOSE PASS many times for each course of action and puts the results side by side. Each run draws Cosmos 2612's habits afresh from the ranges you set, so a plan that only works against one exact opponent shows up as luck.

- **COURSES OF ACTION:** `Do nothing` (always run, as the baseline), `Shadow Cosmos 2612` and `Guard ICEYE-X36`.
- **RUNS:** 10 to 400 for each plan. About 3 seconds a run, shared over your computer's cores.
- **HOURS:** how long each run lasts (6 to 168).
- **The unknowns:** a low and a high value for each of Cosmos's habits.

Press `RUN STUDY`. The table then shows, for each plan: how often it won (with a 95 % range), its average score, how much better it did than doing nothing in the same worlds, fuel used, the closest Cosmos came to X36, and which unknown mattered most.

> **DESKTOP ONLY** A study starts extra copies of the program, which a browser can't do. Run studies from the desktop program.

LIBRARY · SOURCES

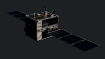
Every reference the numbers rest on, grouped by subject, each with a line on why it was chosen. In game it is under `Help`.

> **REMEMBER**
> The score, the goals and Cosmos's habits are design values. A study tells you which plan does better in this model, not what would happen in real life.

2 · THE MISSION

CLOSE PASS

Monday 26 October 2026, 21:00 UTC. Low Earth orbit, about 540 km up. A Russian inspector satellite is working its way round a US radar satellite. You have two days.



YOUR ORDERS

Cosmos 2612 is closing on ICEYE-X36. Make it leave before it finishes its look. You have no weapons. No war.

In October 2026 the Russian satellite Cosmos 2612 was reported working its way round the commercial radar satellite ICEYE-X36. That really happened, and the game starts from it. What follows is invented: the US sends its own inspector, WATCHMAN-1, to stand beside X36 and make the visitor's job hard.

Cosmos 2612 has a plan. It means to park in three spots near X36, each for six or eight hours, to study it up close: 2 km behind, then 2 km ahead, then 0.8 km behind. When it has finished all three, it goes home with what it came for, and you have lost.

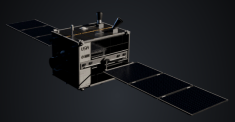
It doesn't want to be watched. It keeps a 5 km bubble round itself, and if WATCHMAN-1 parks inside that bubble while it can see you, it gives way and moves to the far side of X36. Do that enough times and it gives up and leaves.

But it isn't alone either. It reports home by radio every few hours. Every time you crowd it, tension rises in both capitals. Push too hard, too close, and it could end in open conflict, which loses the game outright.

So the mission is a balance: get alongside early, keep a good track on it, crowd it at the right moments, don't go nearer than you need, and spend as little fuel as you can.

You start **20 km behind X36** on the same orbit. Cosmos 2612 is **5.1 km behind X36**, 15 km from you. It has already started.

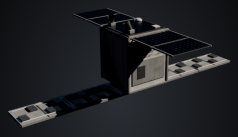
WHO'S WHO



YOU · BLUE · WCH-1

WATCHMAN-1

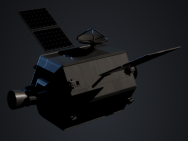
US inspector, invented. 675 kg. Four 22 N thrusters on hydrazine, about 260 m/s of fuel in all. A 0.3 m telescope that sees in visible light and a radio receiver. No weapons, no radar.



PROTECT · GREEN · X36

ICEYE-X36

A real commercial radar satellite that images the ground (ICEYE US). 90 kg with a 3.25 m radar antenna. It has no engine, so it can't get out of the way.



THE INSPECTOR · RED · TRK 4423

COSMOS 2612

A real Russian inspector satellite (NORAD 68762). About 500 kg with only about 90 m/s of fuel. A small infrared telescope it watches you with. It radios home on 2,250 MHz.



YOUR EYES · BLUE · SITES

Five ground radars

Real sites: PFISR in Alaska, LeoLabs in Texas, Arizona and New Zealand, and the Space Fence on Kwajalein. Each sees Cosmos only while it passes overhead.

GOALS AND SCORE

Five goals. Two of them end the game if you fail them. The other three only cost you points. Press F1 in the game to see them ticked off as you play.

GOAL, AS THE GAME WRITES IT	KIND	POINTS	HOW IT IS JUDGED
Make Cosmos 2612 break off before it finishes its look at ICEYE-X36	Must not fail	40	Won when it leaves. Lost if it finishes all three stops. Half marks if it is still at work when the two days end.
Keep Cosmos 2612 more than 1 km from ICEYE-X36	Points	20	Full marks if it never comes inside 1 km. Otherwise (closest ÷ 1 km)^2, so 500 m earns a quarter.
Keep a good track on Cosmos 2612 (within 1 km) at least 80 % of the time	Points	20	Checked every ten minutes: is your track sure to within 1 km, and right? Full marks at 80 % or more.
Use no more than half of WATCHMAN-1's fuel	Points	10	Full marks with a quarter or less used, falling to none with all of it used. Fails past half.
Stay below open conflict (escalation 100)	Must not fail	10	Lost at 100. Points fall with the final escalation: at 30 you keep 7 of the 10.

HOW IT ENDS

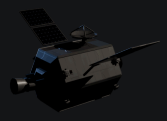


GET ALONGSIDE
Plot move, then Match orbit

→

5 km
CROWD IT
Park that close while it sees you

→



IT LEAVES
The third time, it gives up

OPERATION COMPLETE
Cosmos 2612 gave up, climbed away and left. Your score is out of 100, made of the five goals above.

OPERATION FAILED
It finished its look, or escalation reached 100. Either ends the game at once.

THE 2 DAYS ARE UP
Neither happened in time. You get half the points for the first goal.

A WORKED SCORE

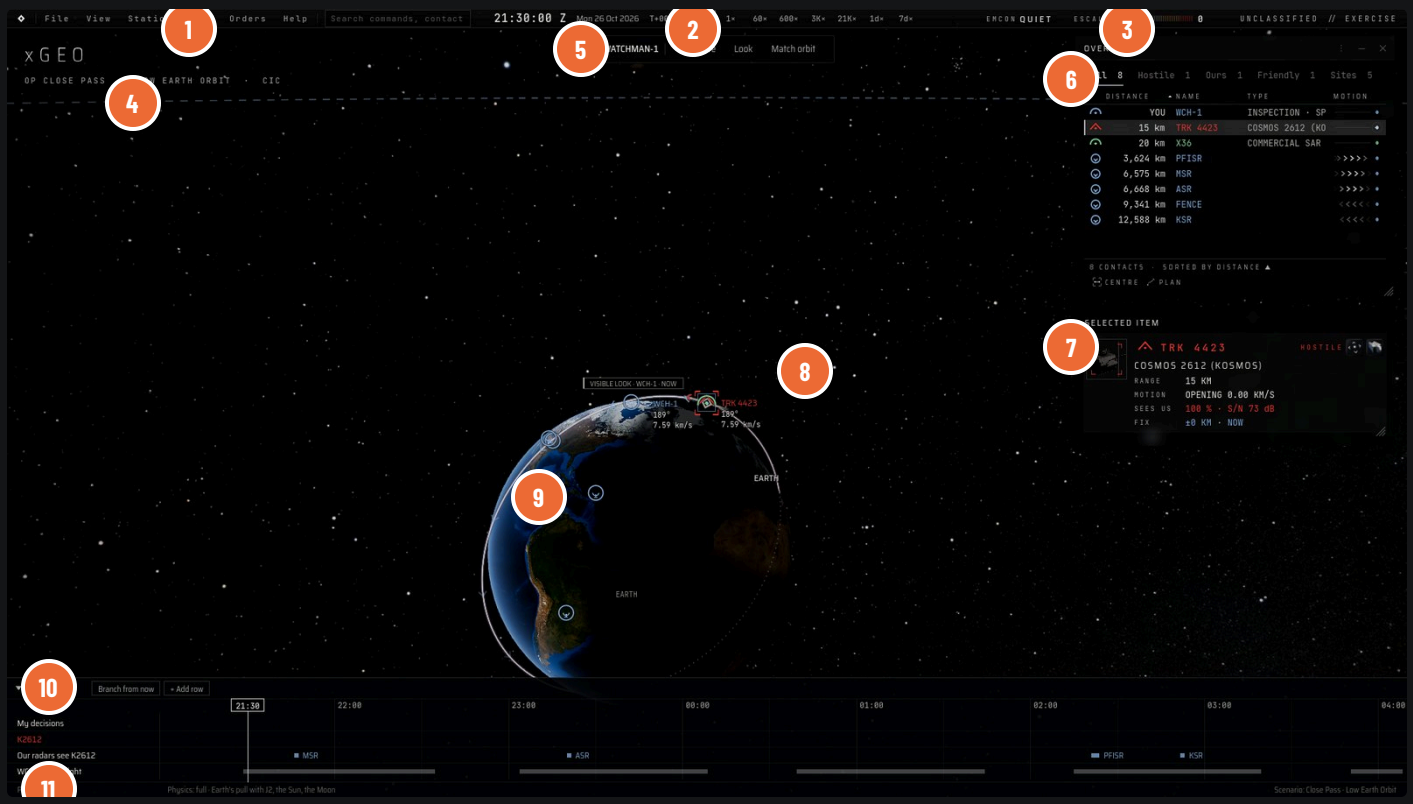
It leaves (40), never inside 1 km of X36 (20), a good track 76 % of the time (19), 30 % of the fuel used (9.3), final escalation 15 (8.5). Total 97.

COLOURS IN THE DEBRIEF

Green from 70 up, amber from 40, red below 40.

ONE SCREEN

Everything happens on one screen: a live 3-D view in the middle, the contact list on the right, the timeline along the bottom. Hover over anything and a card explains it.



1 **MENUS AND SEARCH** File, View, Stations, Map, Orders, Help. Every command, with its key. Type in the search box to find one.

2 **CLOCK** UTC time (Z), the date, mission time (T+), and the time rate, from pause to a week a second.

3 **EMCON AND ESCALATION** Whether you are giving yourself away, and how close the crisis is to war (0 to 100).

4 **OPERATION** OP CLOSE PASS · LOW EARTH ORBIT.

5 **ACTION BAR** The ship you act from and its three orders: Plot move, Look, Match orbit (keys 1 to 3).

6 **OVERVIEW** Every contact, nearest first. Click a row to select it; double-click to centre on it.

7 **SELECTED ITEM** What you know about whatever you clicked: range, motion, whether it can see you, how sure your fix is.

8 **A CONTACT** TRK 4423 is your track on Cosmos 2612. Its label gives course and speed.

9 **THE 3-D VIEW** The Earth, its day and night, your orbit, and the ground radars as rings on the surface.

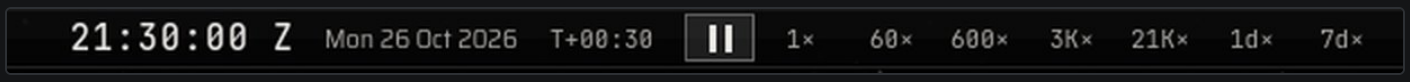
10 **TIMELINE** The next seven hours: your decisions, Cosmos's closest pass, when our radars see it, when you're in sunlight.

11 **STATUS LINE** Paused or running, the physics in use, the scenario. Events show here too (page 12).

3 · THE CONSOLE

THE TOP BAR

A thin bar across the top holds the menus, the search box, the clock, and the two numbers that decide how the crisis is going.



The clock: 21:30:00 Z is UTC. Then the date, the mission time T+00:30, the pause button and the rates.

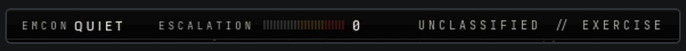
TIME

The game starts paused. Time only moves when you let it. Click a rate, or use `[` and `]` to step slower and faster. **Space** pauses, and from pause it resumes at 600×.

BUTTON	EACH REAL SECOND IS	USE IT FOR
1×	1 second	Watching a burn up close
60×	1 minute	Near the target, listening
600×	10 minutes	Coasting to a meeting
3K×	1 hour	Waiting for Cosmos to move
21K×	6 hours	Long waits, with care
1d×, 7d×	1 day, 1 week	Not useful in a 2-day mission

One trip round the Earth takes about 95 minutes. At high rates your computer may not keep up: the game then runs as fast as it can and the clock shows the real rate.

EMCON, ESCALATION, CLASSIFICATION



EMCON (emission control) reads `QUIET` while you only listen and `LOUD` while any of ours sends a radar beam or jams. WATCHMAN-1 has neither, so in CLOSE PASS you are always quiet. Being quiet matters: if Cosmos hears you, it goes silent.

ESCALATION is a bar of 24 cells and a number from 0 to 100: how close the two sides are to open conflict. At 100 you lose. Chapter 12 lists exactly what raises it.

`UNCLASSIFIED // EXERCISE` reminds you this is a training scenario: invented events on real physics.

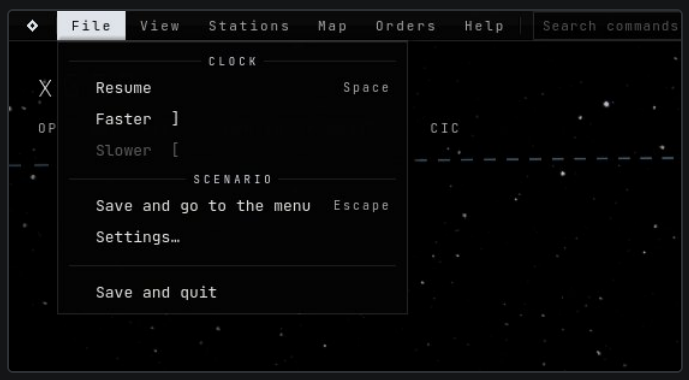
SEARCH



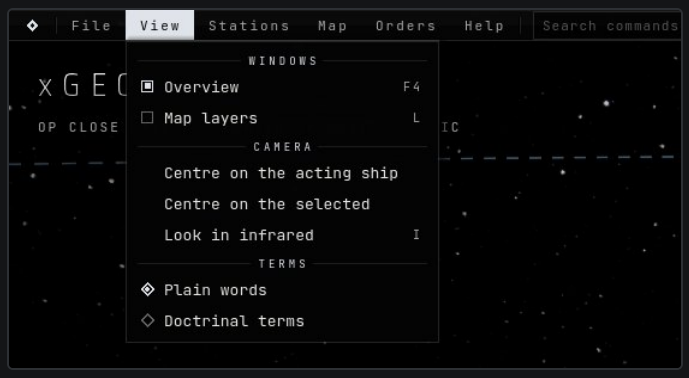
Type part of any command into `Search commands, contacts…`. Up to 12 matches appear as `Menu › Item`. **Enter** runs the first, **Esc** clears it.

THE MENUS

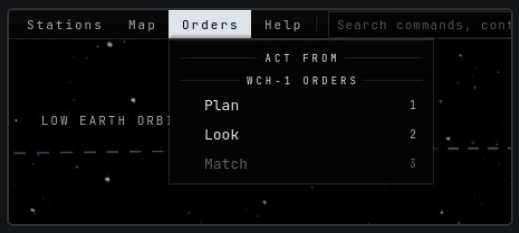
Every command in the game is in one of six menus, with its key beside it. Hover another title while one is open to switch; click elsewhere or press Esc to close.



File. Pause or resume, faster, slower. Save and go to the menu (Esc), Settings, Save and quit.



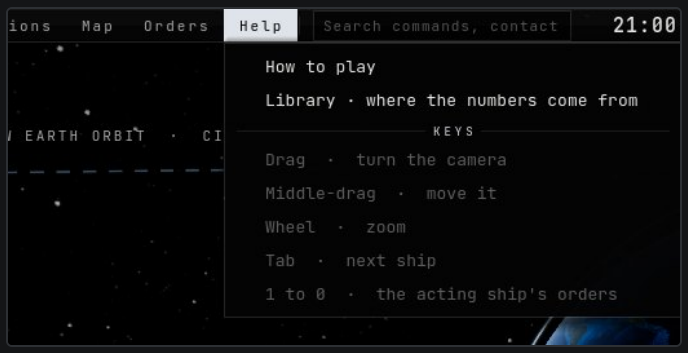
View. Show the Overview (F4) and Map layers (L). Centre the camera on your ship or the selection. Look through the telescope (I). Terms: plain words or doctrinal terms.



Orders. The acting ship's orders with their keys: Plan 1 , Look 2 , Match 3 . Greyed when not possible now.



Stations. The windows: Campaign F1 , SIGINT station F2 , Signals, Planner F3 , Database F5 , Heat forecast F7 , Solar system. A tick marks each one that is open.



Help. How to play, the Library, and a reminder of the mouse keys.

MAP

The same list as the Map layers panel (page 14): tick what the 3-D view draws, or go `Back to the defaults`.

TERMS

Plain words and doctrinal terms change only the words, never the game. Some pairs:

PLAIN	DOCTRINAL
Listening station	EW console
Plan a move	Maneuver planner
Find direction · Identify signal	DF · Classify
Fuel	Δv
What you must do	Objectives
How it went	After-action review

With doctrinal terms on, hover cards also show the formula behind a number.

OVERVIEW AND SELECTED

Top right: every contact you know of, and under it everything you know about the one you picked.



OVERVIEW F4

Tabs, each with a count: `All`, `Hostile` (Cosmos), `Ours` (WATCHMAN-1), `Friendly` (X36, which you protect) and `Sites` (the five radars).

Columns: `DISTANCE` from your ship (your own row says `YOU` and stays on top), `NAME`, `TYPE` and `MOTION`. Click a column title to sort; click again to reverse.

MOTION: chevrons flow inward while a contact closes on you and outward while it opens, faster when quicker. A beating dot means a sensor holds it right now; a hollow ring that fades means it hasn't been seen for a while. The whole row fades as a track gets old.

Click a row to select it, double-click to centre the camera on it, right-click for its orders. `Centre` and `Plan` at the foot act on the selection.



SELECTED ITEM: A CONTACT

RANGE	How far it is from your ship.
MOTION	`CLOSING` or `OPENING`, and how fast, in km/s.
SEES US	Its telescope's chance of seeing you on each look, and the signal-to-noise. Red when it would see you.
FIX	How sure you are of where it is (±, at 2σ), and `NOW` or how old the last sighting is.



SELECTED ITEM: ONE OF OURS

HEIGHT · SPEED	Above the surface, and orbital speed.
ΔV LEFT	Fuel, as the speed change you can still make.
DOING	`Coasting`, or `Burn set for 22:40Z`.
THREAT	The nearest hostile and how far it is.
ORBIT	Lowest × highest height, and tilt.
STATE	Lit or in shadow, until when, and Quiet or Loud.

Click the name to open its Database page.

THE TIMELINE AND EVENTS

The Timeline along the bottom is your look ahead: the next seven hours, on the same clock as everything else.



Four rows to begin with. The boxed 21:30 is now; the grid is every half hour.

THE ROWS

My decisions	Burns you have set, and the move the planner has open (`Move WCH-1 by 21:00`).
K2612	The nearest hostile: its closest pass to anything, as `Closest · 2 km · X36 · 21:53`.
Our radars see K2612	A bar for each radar pass, named PFISR, MSR, KSR, ASR or FENCE.
WCH-1 in sunlight	Bars while you're lit. The gaps are Earth's shadow.

+ ADD ROW

Add up to nine rows, from four groups:

- **Platforms:** any craft's distance from you, height, or sunlight.
- **Sensors:** when each radar sees each craft, and when Red's telescopes see you.
- **Conditions:** when you are within 1, 5, 10, 25 or 100 km of Cosmos; when a craft is in shadow.
- **Orbits:** equator crossings, highest and lowest points, height.

Hover a row's name for an `×` to remove it. Click the arrow by TIMELINE to fold it away.

BRANCH FROM NOW

Try something without losing what you have. `Branch from now` keeps the present as Run 1 and plays on as Run 2. Tabs appear on the title line; click one to go back to that run, exactly where you left it.

A GOOD USE

Branch before a big move. If Cosmos reacts badly, go back to Run 1 and try a smaller one. Branches are not saved to disk.

EVENTS



Bottom left. Folded, it shows the newest event and how many are new; click to see the last four. Each has its mission time and a class:

WARNING	Act now: a burn seen, tension near a limit.
CAUTION	Worth knowing: lost sight of a track.
NOMINAL	Good news: it gave way.
ADVISORY	For the record: it parked, you burned.

MOVING AROUND

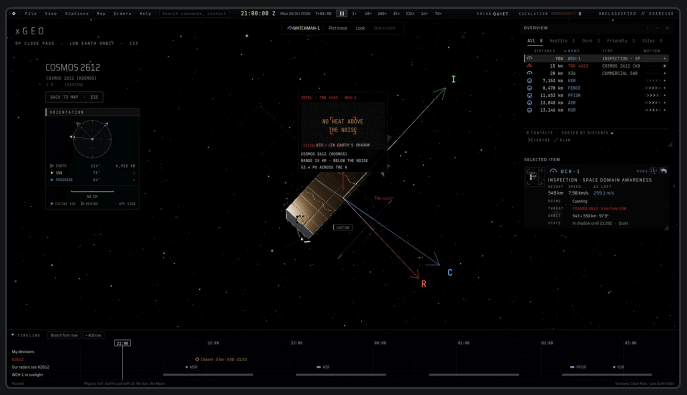
The view is a real 3-D picture of the Earth and everything near it, from 4 metres away to beyond the planets. The camera always turns round a focus: your ship to begin with.

DO THIS	TO
Left-drag	Turn round the focus
Right-drag or middle-drag	Slide the view off the focus (Shift + left-drag works too)
Wheel	Zoom in and out
W S	Turn over and under
A D	Turn around
Q E	Zoom out and in
Hold Shift	Keys three times faster
Click	Select a contact
Double-click	Focus on a craft, the Earth or the Moon. Ours and identified craft come in close
F	Close-up on the selection
Tab	Focus on the next object
Right-click	Orders for what is under the pointer (page 15)

When several objects sit on top of each other, a left click opens a small SELECT list to pick one. Camera keys do nothing while you type in a box or a station is open.



Close-up (F) on WATCHMAN-1: the real model, lit by the real Sun, with its axes. Esc or a double-click on empty space backs out.



Cosmos 2612 up close. You can only come in on a craft you have identified.

CONTACTS AND LAYERS

Every craft on the map is drawn where *you believe* it is. For your own ships that is the truth. For Cosmos 2612 it is your track: a best guess with a size.



WCH-1 and TRK 4423, 15 km apart. The plate `VISIBLE LOOK · WCH-1 · NOW` marks a fresh telescope sighting.

COLOURS AND SYMBOLS

Blue · ours Green · protected Red · hostile

Yellow · unknown Grey · neutral

Ours and friendly craft wear an arc over a dot; a hostile, a pointed roof; an unknown, an open box. Ground radars are circles on the Earth. In Map layers you can switch to the military MIL-STD-2525D frames instead.

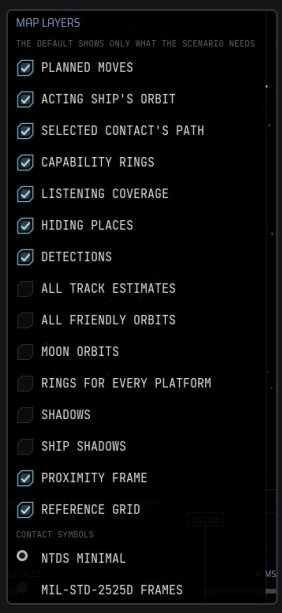
Hovered or selected, a contact shows three lines: name, course and speed. Selected, a hostile also shows its **uncertainty region**: the space it could be in, drawn at 2σ, a size that holds it most of the time.

WHEN YOU LOSE SIGHT OF IT

A contact you can no longer see stays at its **last known position**, drawn dashed and fading, with an age ring that fills clockwise and a chip such as `LAST 2H 14M`. Near the end it reads `DROPS IN 3H 10M`, then `ABOUT TO DROP`.

A track is **dropped** after 24 hours unseen, or once it is so unsure it could be anywhere. Seen again, its icon jumps to the new fix with a short ring.

Recent sightings show as pings for half an hour: telescope lines, radio bearings and radar fixes.



MAP LAYERS L

Tick what the view draws. The ones that matter in CLOSE PASS:

Planned moves	Your transfer arcs, with hour marks
Acting ship's orbit	Where you coast next
Selected contact's path	Where it goes next (Red: your estimate)
Detections	Last half hour's sightings
All friendly orbits	X36's orbit too
Proximity frame	Close up: the R, I and C axes and the other ship's path
Ship shadows	Hull shadows up close; costs frame rate

Your choice is remembered next time.

5 · ORDERS

THE ACTION BAR

WATCHMAN-1 has three orders, on the bar at the top centre and on the number keys. None of them is a weapon. Presence is the tool.





PLOT MOVE

Opens the planner on the selected contact (same as F3). Work out when to burn and what it costs.

USES FUEL



LOOK

The telescope view of the selection (same as I). Far off a point of light, close in the hull itself.

QUIET



MATCH ORBIT

One burn now that puts you on the selected craft's orbit where you are, so you coast alongside it.

USES FUEL

MATCH ORBIT, IN DETAIL

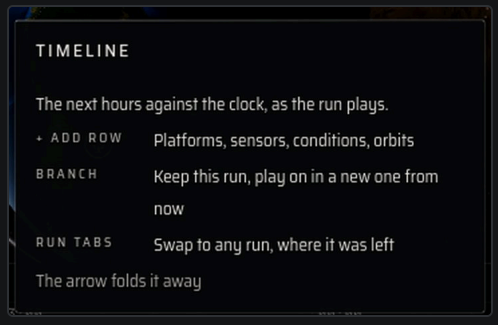
Arriving isn't stopping. A transfer brings you to the target, but you fly past at a few tenths of a metre a second. **Match orbit** cancels that difference, so you stay beside it.

It needs a contact selected that is within **5,000 km**, and enough fuel. Its hover card ticks off each condition and shows the cost. Matching Cosmos uses your *track's* idea of its speed, so a good track gives a better match.

GREY MEANS NOT NOW

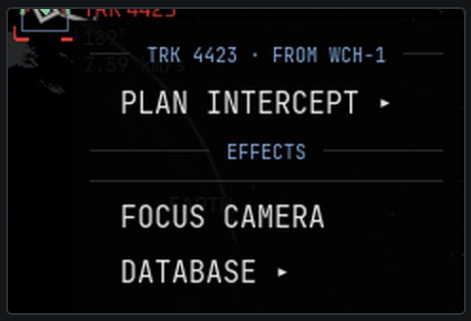
An order that can't be done is greyed out. Hover over it and the card says why, for example `Get within 5,000 km first: now 6,120 km`. The name at the start of the bar is the ship you act from; click it to centre on it.

HOVER CARDS



Rest the pointer on anything for a third of a second and a card explains it: a title, short rows, ticks and crosses for conditions, amber for prices, and the keys.

RIGHT-CLICK A CONTACT



PLAN INTERCEPT ▸	Opens the planner on it
FOCUS CAMERA	Turns the view round it
DATABASE ▸	Its page in the Database (F5)

The menu also has an `EFFECTS` heading for counterspace effects such as jamming. WATCHMAN-1 carries none, so in CLOSE PASS the heading stays empty.

Right-click empty space for `FOCUS WCH-1` and `MAP LAYERS…`.

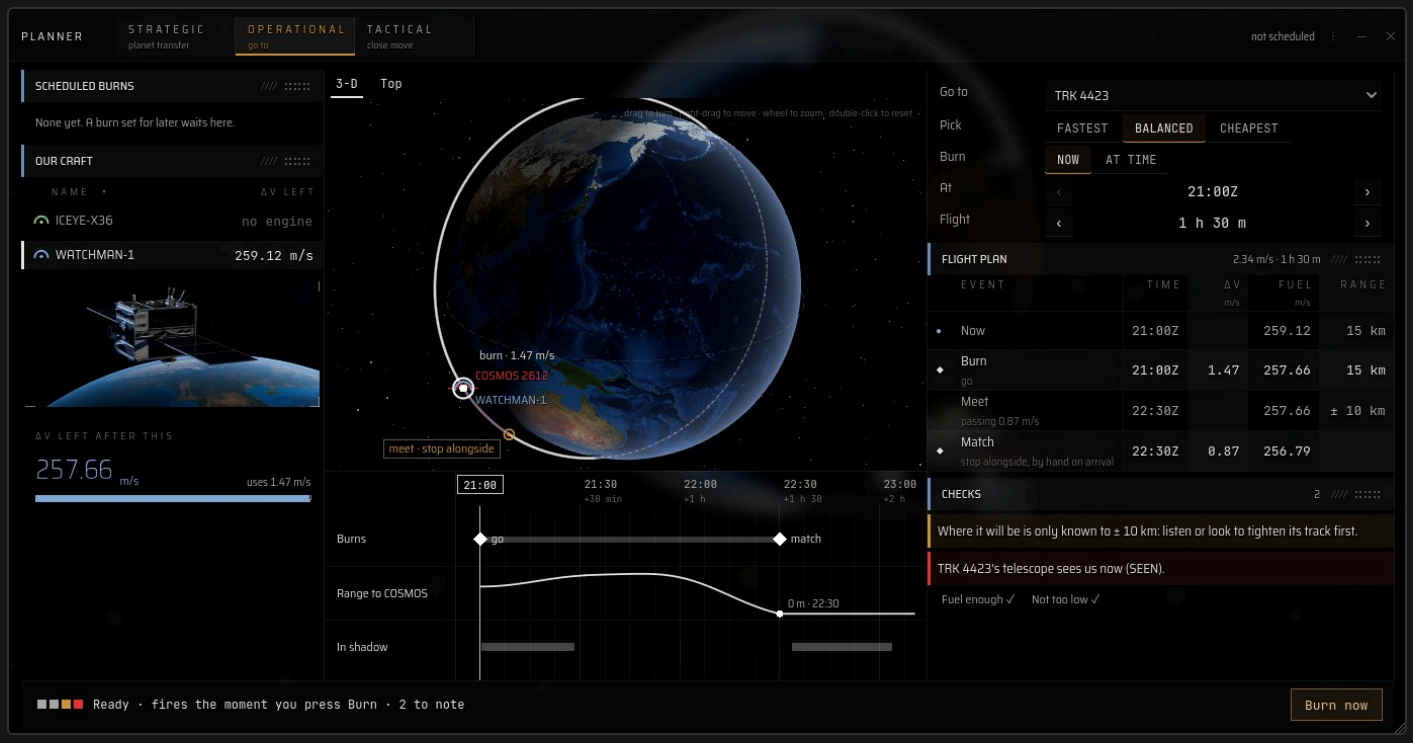
BURNS ARE SEEN

Every burn makes your thrusters glow. Cosmos's telescope can catch it, and if it sees one of ours burning it goes quiet on the radio for six hours. Listen first, then move.

6 · THE PLANNER

PLAN A MOVE

The planner does the orbital maths. You say where to go; it solves every way there over the next day, under full gravity, and offers the best ones with their price.



Going to TRK 4423, BALANCED: burn now for 1.47 m/s, fly 1 h 30 m, then 0.87 m/s to stop alongside. 2.34 m/s in all, leaving 256.79 m/s.

OPENING IT

Select a contact and press `1` `Plot move` or `F3`, or right-click it and choose `PLAN INTERCEPT`. With your own ship selected it opens asking where to go.

THREE TABS

STRATEGIC	A trip to another planet (page 20).
OPERATIONAL	Go to a craft, an orbit, the Moon or an Earth–Moon Lagrange point. This is where most moves start.
TACTICAL	A close move: one small push, fired now, in the target's own frame (page 19).

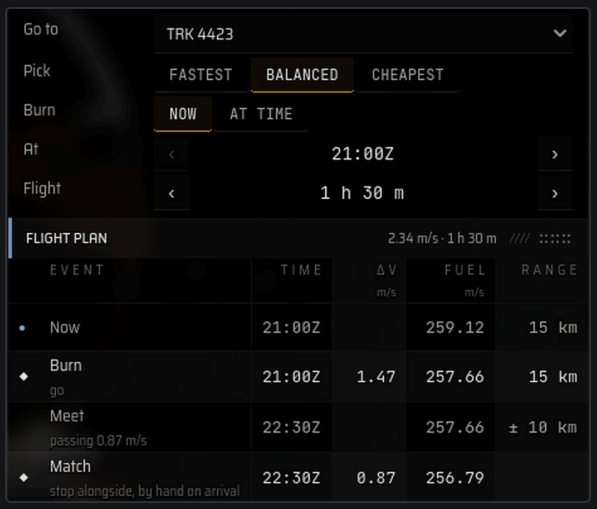
THE PARTS

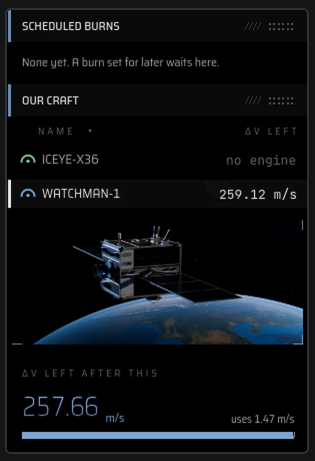
- **Left:** `SCHEDULED BURNS` waiting to fire; `OUR CRAFT` (click one to plan for it); `ΔV LEFT AFTER THIS`, green if you can afford it, with a bar of what this plan uses.
- **Centre:** the picture, in `3-D` or `Top` view (drag to turn, wheel to zoom, double-click to reset), and a timeline under it.
- **Right:** the form, the `FLIGHT PLAN` table and the `CHECKS`.
- **Foot:** one lamp per check, the ready line, and `Burn now`.

Any open plan also shows on the main map as `PLANNED MEETING`, with how long until you get there.

OPERATIONAL: GOING SOMEWHERE

Pick a destination, pick a trade between time and fuel, and choose when to burn.





ROW	WHAT IT DOES
Go to	A list: every craft you know of, preset orbits, the orbit another craft is on, the Moon, and Earth–Moon L1 to L5.
Pick	`FASTEST` gets there soonest you can afford. `CHEAPEST` spends the least, taking longer. `BALANCED` (the default) is a fair trade. Cheapest and Balanced count the stop at the end too.
Burn	`NOW` or `AT TIME`.
At	Step through the burn times the planner solved (‹ ›).
Flight	Step through the flight times (‹ ›).

Behind the picks is a grid of burn times (now, then every 1½ hours for a day) against flight times (1 to 6 hours for a nearby craft). `At` and `Flight` walk it by hand.

THE FLIGHT PLAN

Each row is an event, with its time (UTC), its Δv, the fuel left after it, and the range to the target:

- **Now**: where you are.
- **Burn · go**: the first burn.
- **Meet · passing 0.87 m/s**: you arrive, still moving that fast past it. Range is how sure you are of where it will be.
- **Match · stop alongside, by hand on arrival**: the second burn. **You press Match orbit yourself** when you get there.

Going to an *orbit*, the Moon or a Lagrange point, the arrival burn is automatic: the engine schedules it for you.

PRESET ORBITS

GO TO	LOW × HIGH	TILT
Low, circular	500 km	yours
Sun-synchronous	700 km	about 98°
Medium (GPS)	20,180 km	55°
Transfer to GEO	250 × 35,786 km	27°
Geostationary · Above GEO	35,786 · 36,086 km	0°
Molniya · Tundra	600 × 39,700 · 24,400 × 47,100	63.4°

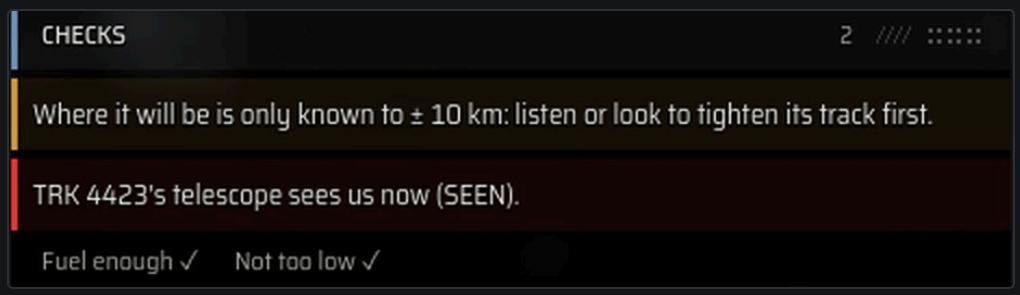
Each preset adds `Low · high` and `Tilt · node` steppers so you can describe any orbit.
`Orbit · the one X36 is on` joins another craft's orbit without meeting it.

WITH 260 M/S

WATCHMAN-1 can't reach GPS, GEO or the Moon. The planner still draws the cheapest way, dashed in amber, with its price. It is a good way to see how expensive space is: going from low orbit to GEO costs at least 4 km/s, fifteen times what you carry.

CHECKS AND THE PICTURE

Before you burn, the planner checks the plan and draws what it will do. Read both: they tell you what could go wrong.



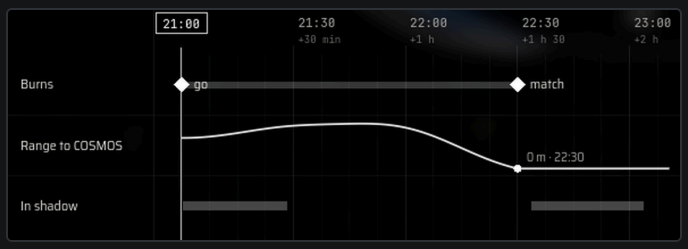
CHECK	COLOUR	WHAT IT MEANS
Fuel enough	red	You can't afford the burn at all.
Fuel to stop	amber	You can get there but not stop. Plan cheaper.
Not too low	red	The path dips below 122 km, where the air would bring you down.
Arrive ± 10 km	amber	Where the target will be is only known to that size. Listen or look to tighten the track first. Under 5 km it passes.
Seen COLD · WARM · SEEN	clear · amber · red	Whether Cosmos's telescope can see you now, and when that next changes (page 24).

THE PICTURE

Soft blue	Your orbit now
White	The planned path
Dashed sand	The orbit you're going to
Dashed amber	A way you can't afford
White ring, red ticks	The burn point and its size
Amber ring	The meeting point

Faint dots show where the Earth hides a line. Orbits are drawn at their true tilt.

THE TIMELINE UNDER IT



Filled diamonds are burns (◆ go, ◆ match); outlines are automatic steps. Going to a craft you see the range falling to `0 m` at the meeting, and when you're in shadow on the way.

BURN

`Ready · fires the moment you press Burn` means go. `Burn now` flies it at once. Set a later time and the button reads `Burn at 22:30Z`: the burn waits in SCHEDULED BURNS and fires by itself.

TACTICAL: CLOSE MOVES

Up close, you don't plan transfers. You give a small push and watch where it carries you over the next two orbits, drawn round the target.



A push of +0.50 m/s along the orbit, drawn round Cosmos 2612. The amber label marks the closest pass: 14 km at 21:11Z.



Set a push in three directions, each with ‹ › buttons. `Step` picks how much each click adds: 0.01, 0.1 (the default) or 1 m/s. `Clear` zeroes it. `Burn now` fires it at once; a close move can't be scheduled.

Radial (R)	Away from the Earth (+) or toward it (–).
In-track (I)	Along the orbit (+) or against it (–).
Cross-track (C)	Sideways, along the orbit's pole.

WHAT EACH PUSH DOES

Orbits don't behave like cars. These three rules are all you need:

- **Push forward (+I) and you drift behind.** The extra speed raises the far side of your orbit; a higher orbit is slower, so you fall back. Push backward (–I) to drift ahead.
- **A radial push makes a loop.** +R or –R swings you in an oval round where you were, once an orbit, with no drift. Good for circling a target.
- **A cross-track push swings you side to side**, crossing back once an orbit.

The `WHAT IT DOES` table gives the push, the closest pass, and when you next enter shadow and sunlight. The timeline adds `Can COSMOS see us?` with its telescope.

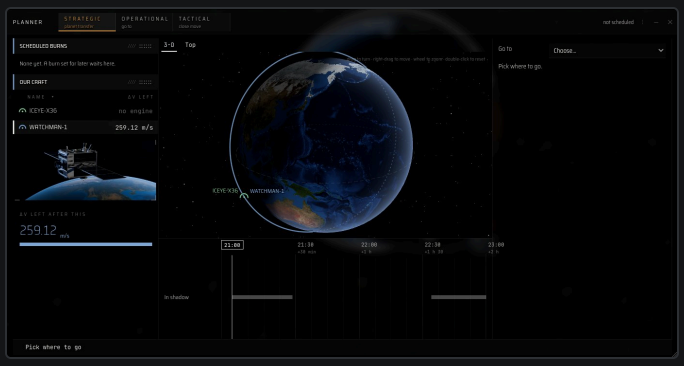
The tab turns on the Proximity frame layer and brings the main camera in close.

> **RULE OF THUMB** In low orbit, 0.1 m/s along the track moves you roughly 1.7 km per orbit. Tiny pushes, patiently, do the work.

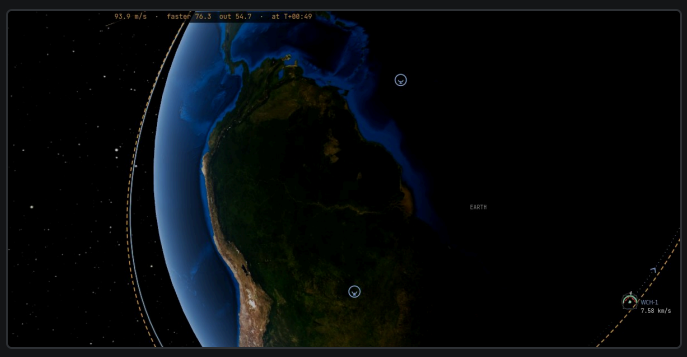
6 · THE PLANNER

FURTHER, AND BY HAND

The planner reaches the Moon and the planets too, and you can always plan a burn by grabbing your own orbit and dragging it.



STRATEGIC: a trip to another planet. Its timeline row `Cost to leave` runs from teal (cheap) to amber (dear) by departure date.



Dragging the orbit: the new orbit swings out in dashed amber, and the box gives the burn: its size, how much faster or slower, how much out or in, and when.

STRATEGIC

`Go to` any planet. Set the orbit to arrive in (`Arrive in` and `Tilt`), when to leave, and the flight time, in days. Speeds are in km/s. The engine corrects the burn under full gravity on the way, aims at the low point, and brakes into orbit by itself.

THE MOON AND THE LAGRANGE POINTS

In OPERATIONAL, `The Moon · an orbit round it` picks a height (100 km to begin) and tilt. `Earth–Moon L1` to `L5` are the five balance points. The picture centres on the goal, shows the Moon's month-long path, and how far you would drift from a point in a week. L1 to L3 are unstable; L4 and L5 are stable.

WATCHMAN-1 can't afford any of these. They are there because the engine can.

DRAG AN ORBIT

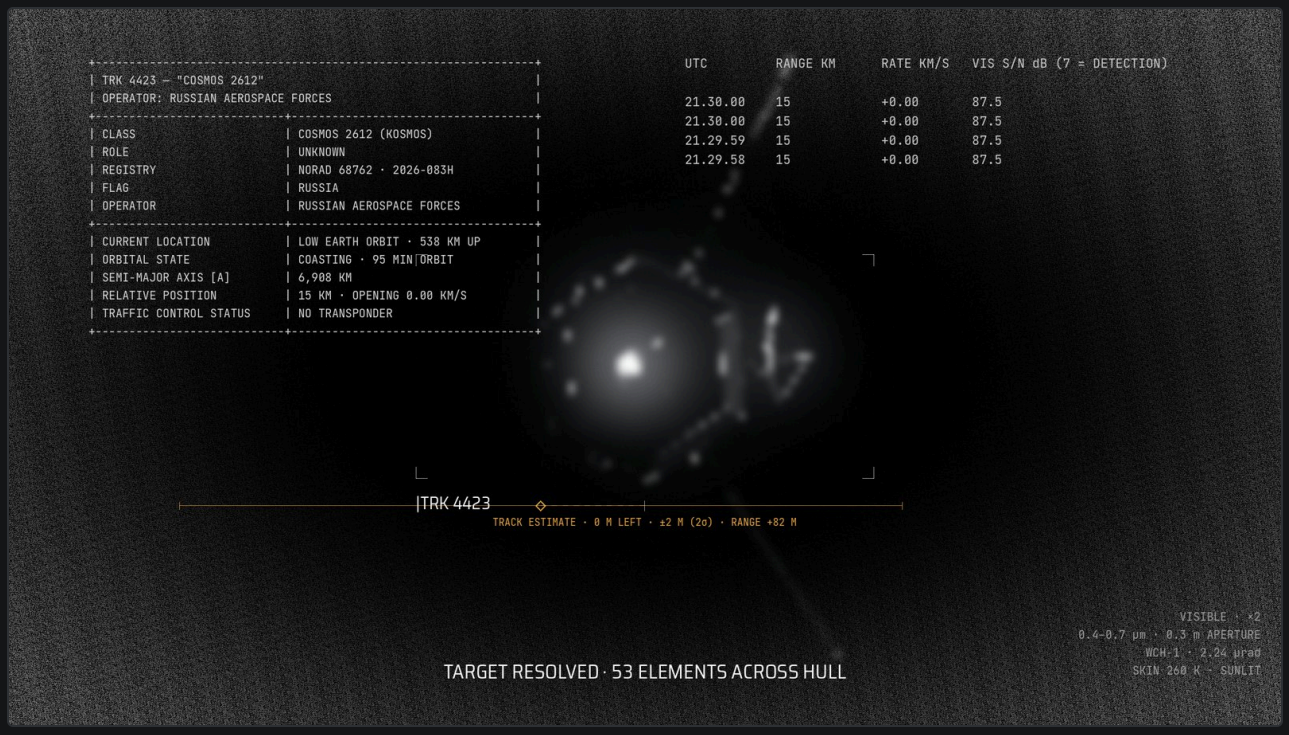
1. Press the left button **on your own orbit line**, at the point where you want to burn.
2. Drag **along** the line to speed up (forward) or slow down (back), and **across** it to push out or in. Each pixel is 0.5 m/s.
3. Watch the new orbit and the readout, such as `2.8 m/s · faster 2.1 out 1.8 · at T+00:42`.
4. Let go. The burn is set for the next time you pass that point, and the log says so.

Right-click or **Esc** while dragging cancels. Under 0.1 m/s does nothing. Holding **Shift** stops a drag starting, so you can turn the view instead. The preview is simple two-body motion; the burn itself flies under full gravity.

7 · THE TELESCOPE

LOOK

WATCHMAN-1's 0.3 m telescope is how you see Cosmos 2612 for yourself. From far off it is a point of light. Close in, it is a hull you can name.



TRK 4423 at 15 km: `TARGET RESOLVED · 53 ELEMENTS ACROSS HULL`. Top left, the data table; top right, the reading log; the bracket marks where your track says it is.

USING IT

Select a contact and press 2 Look or I. V switches between visible light and infrared on ships that have both (WATCHMAN-1 has visible only). Esc closes it.

WHAT THE CAPTION TELLS YOU

TARGET RESOLVED	You see the hull, so many pixels across.
UNRESOLVED · POINT SOURCE	A dot. Seen, but too far to show shape.
BELOW DETECTION	Too faint to pick out.
IT IS IN EARTH'S SHADOW	Nothing to see in visible light. Wait for the Sun.
NO LINE OF SIGHT	The Earth is in the way.
DRIVE BURNING	You caught it firing its thrusters.

THE TELESCOPE WORKS FOR YOU ALL THE TIME

You don't need the view open. Every 10 minutes WATCHMAN-1's telescope looks at each contact it tracks, and each sighting tightens the track. But it can only see a target that is:

- **in sunlight**: in Earth's shadow there is nothing to reflect;
- **not against the Earth's bright face**, or close to its edge;
- **bright enough** for its distance.

Close in, the telescope also measures **range** from how big the hull looks: to about 30 m at 5 km, 0.5 km at 20 km, nothing useful past about 160 km. It can name a hull inside about 60 km.

WHY THIS MATTERS

The ground radars see Cosmos only on their passes. In between, your telescope keeps the track good, but only on the sunlit part of each orbit. That is why the custody goal rewards staying near.

8 · THE SIGINT STATION

LISTENING

Cosmos 2612 reports home in short radio bursts. The SIGINT station (F2) is your receiver: it shows each burst arriving live, records it, and lets you work out who sent it.



A burst arriving: the bright stripe on the waterfall and the peak on the spectrum at 2,250 MHz, in S band, while the recording list shows it `ON AIR`.

ACROSS THE TOP

EAR	The receiver you listen from. In CLOSE PASS, only WCH-1.
SPAN	How much time the waterfall shows: 30 M, 2 H or 12 H.
EMCON	QUIET or LOUD.
HEARS ITS REPORTS TO	How far away this ear hears a report half the time, worked out live.

BANDS

S BAND	2200–2290 MHz · telemetry and control. **Cosmos 2612 talks here, at 2,250 MHz.**
X BAND	8400–8450 MHz · command links
KA BAND	31800–32300 MHz · datalinks

Each band has a small live waterfall under its button. B steps to the next band. The station jumps to the band of the newest recording.

WATERFALL AND SPECTRUM

The **waterfall** is frequency across and time down, newest at the top. A burst is a bright dash; one still on the air has a pulsing amber cap. Hover for its frequency and time; click to tune to it.

The **spectrum** under it is the receiver right now (the white `LIVE` trace) over a one-hour peak hold. The red dashed line `EVEN ODDS 6.8 dB` is where a burst is heard half the time.

WHEN IT TALKS

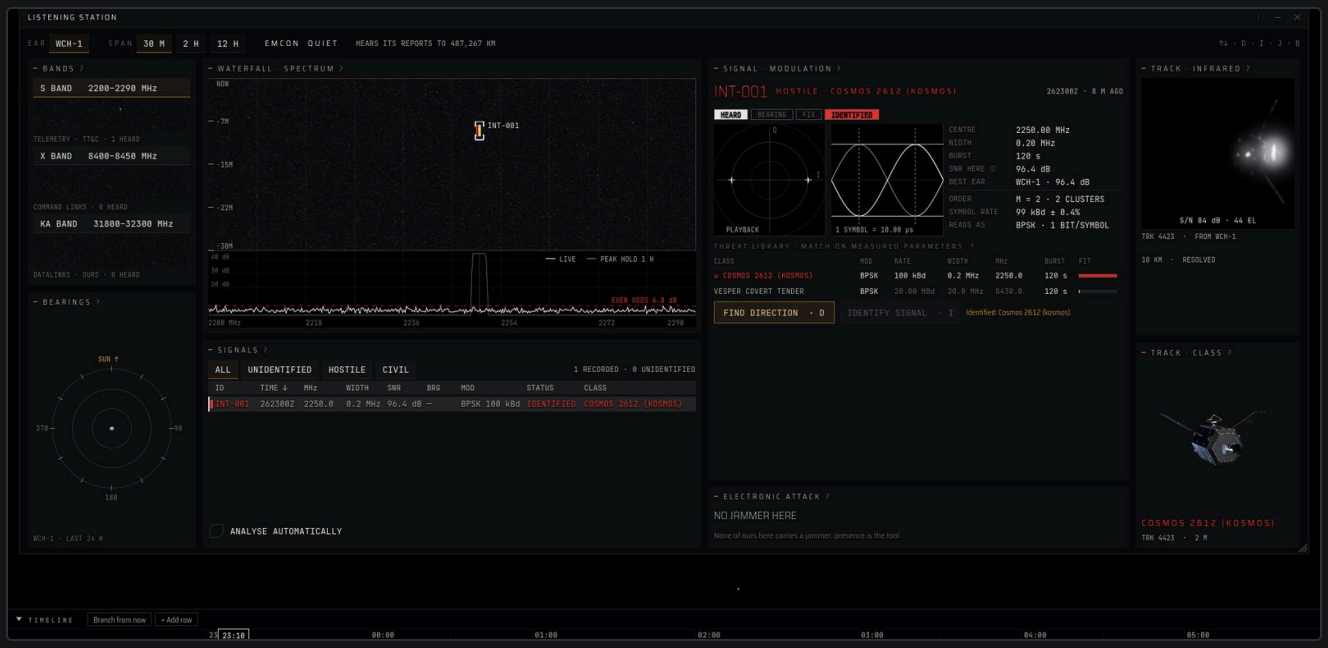
- A first report within the first three hours, then every **3 to 6 hours**, and one after each move. Each lasts **2 minutes**.
- Only while its ground station near Moscow can see it, so reports bunch on passes over Russia.
- It goes **quiet for 6 hours** if it sees one of ours burning, hears us, or one of ours is within **10 km**.

> **SO** Listen before you close in. Once you're within 10 km it stops talking.

8 · THE SIGINT STATION

WORKING A SIGNAL

A recording is a clue. Measure it, find where it came from, and match it to a known radio, and you know who is talking.



INT-001 worked: the four-dot constellation and eye diagram read as BPSK at 100 kBd, the threat library matches it to Cosmos 2612, and the status reads IDENTIFIED.

THE RECORDINGS LIST

Filters: `ALL`, `UNIDENTIFIED`, `HOSTILE`, `CIVIL`. Columns: `ID` (INT-001…), `TIME`, `MHz`, `WIDTH`, `SNR` (strength), `BRG` (bearing), `MOD`, `STATUS` and `CLASS`. `↑` `↓` step through it.

Status goes `HEARD` → `BEARING` → `FIX` → `IDENTIFIED`. Recordings stay in the list all game, but you can only work one for **1 hour** after the burst ends.

READING THE SIGNAL

Count the dot clusters in the **constellation**: 2 or 4 is BPSK or QPSK, 8 is 8PSK. The **eye diagram**'s gap is one symbol. The readouts give centre, width, burst length, `SNR HERE ⓘ`, order, symbol rate and `READS AS`.

The **threat library** lists known radios, best fit first, with what agrees shown bright.

THE TWO BUTTONS

FIND DIRECTION · `D`	Takes a bearing from every ear that heard it. One ear gives a line; two ears far apart cross to a place. With one ear, a bearing still tightens a track you already have.
IDENTIFY SIGNAL · `I`	Names its class and side. Needs the burst to have ended, and **at least 15 dB** at the best ear. No bearing needed first.

Too weak? The hint says so: wait for a stronger report or get closer (but not within 10 km).

`ANALYSE AUTOMATICALLY` works every recording as it arrives. Handy, but you learn nothing.

Click `SNR HERE ⓘ` to see the whole link budget, term by term (page 26).

SIGNALS WINDOW

`Stations › Signals`: a small window with the six newest recordings. Click one to open the station on it.

THE HEAT FORECAST

Cosmos 2612 watches you too, with a small infrared telescope. The heat forecast (`F7`) shows how plainly it sees you now and over the coming orbit.



The forecast at the start: `SEEN`. TRK 4423 sees you best, at 15 km. Left, your orbit from above the North Pole, coloured by how visible you'll be, with the Earth's shadow and the Sun.

WORD	MEANS
COLD	Hidden. Its chance of seeing you on a look is 2 % or less.
WARM	Seen now and then.
SEEN	Seen on most looks.
—	No Red telescope is known.

The right side gives `YOUR HEAT NOW`, a meter from HIDDEN to SEEN, and three rows: `NEXT SHADOW`, `QUIET BURN` (the best moment to burn unseen) and `HIDDEN FOR`.

It looks one orbit ahead (at least 3 hours), assumes everyone coasts, and places Red's telescope where *your track* says it is.

WHY BEING SEEN MATTERS HERE

In most games you want to hide. In CLOSE PASS you mostly want the opposite: **crowding only works if Cosmos can see you.** When you park within 5 km, check `SEES US` in the Selected item. If it reads 0 %, wait for a better angle.

Being seen has a cost too. When it sees you burning, it goes quiet on the radio for six hours, so you hear nothing.

GOING DARK

The `GO DARK` button stores a ship's waste heat in a heat sink so its radiators go cold. WATCHMAN-1 has no radiators and no sink, so it is greyed:

```
Go dark: no radiators to hide. The hull alone shows.
```

`PATH ON MAP` draws the forecast on the 3-D view in the same colours.

STREAKS

Hours spent COLD without a break earn ranks: QUIET at 1 h, DARK at 6 h, BLACK at 24 h, GHOST at 72 h.

TRACKS AND RADARS

You never see Cosmos 2612 directly. You see your *track* on it: a best guess, built from every sighting, that grows less sure with every minute it isn't seen.

A TRACK IS A BELIEF

Cosmos 2612 is in the public catalogue, so you start with a track about 200 m off, already named. From then on:

- **Each sighting tightens it.** Sensors look every 10 minutes. Radar fixes come every 10 seconds during a pass.
- **Between sightings it widens.** The track coasts on full gravity, but small unknowns grow.
- **A bearing alone can't tell range.** Only radar and the telescope's hull size do.
- **Unseen burns.** When a sighting lands far from where the track expected, the tracker decides Cosmos burned without being seen and widens the track enough to explain it (1, 5, 20 or 50 m/s).
- **Seen burns are sized.** If your telescope catches its thrusters, you get `TRK 4423 just burned. It's on the move.` Later fixes size it: `TRK 4423 burned about 0.6 m/s (give or take 0.1).`

CUSTODY

The custody goal counts a track as good when it is sure to within 1 km **and** right to within 1 km. You need that 80 % of the time. `FIX ±0 KM · NOW` in the Selected item is the quickest check.

THE FIVE RADARS

All always on. Each fixes a craft only while it is above the radar's lowest angle and within its reach. The Timeline row shows each pass.

NAME	WHERE	LOWEST · REACH
PFISR	Poker Flat, Alaska · UHF	45° · 3,000 km
MSR	LeoLabs Midland, Texas · UHF	30° · 3,000 km
ASR	LeoLabs Arizona · UHF	20° · 3,000 km
KSR	LeoLabs Kiwi, New Zealand · S band	40° · 1,500 km
FENCE	Space Fence, Kwajalein · S band	20° · 3,000 km

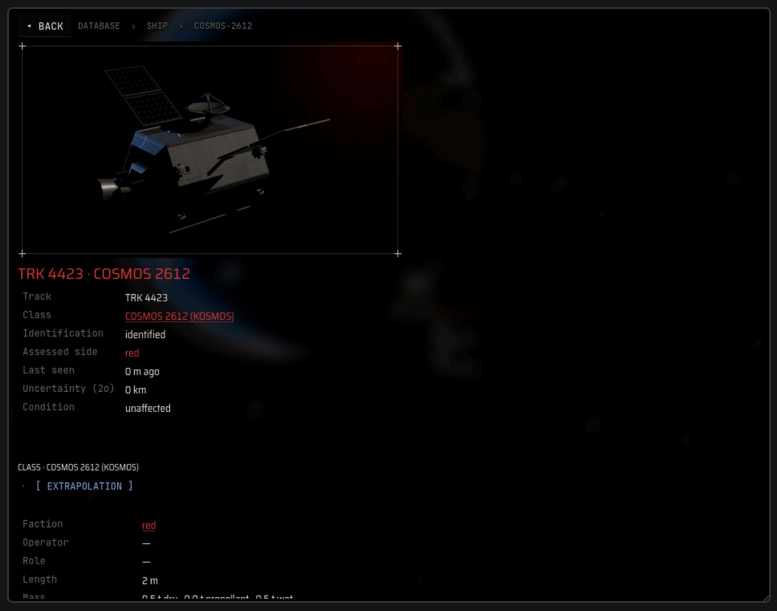
Each measures range to about 15 m and angles to a few hundredths of a degree. A pass lasts a few minutes; there can be hours between them.

LOST, THEN DROPPED

In contact	Seen on the last look. `NOW`, a beating dot.
Out of contact	Not seen: `Lost sight of TRK 4423.` Shown dashed at its last known position.
Dropped	Unseen for 24 hours, or too unsure to say where it is.

DATABASE, CAMPAIGN, SUMS

Three windows tell you what things are, how the mission stands, and how a number was worked out.



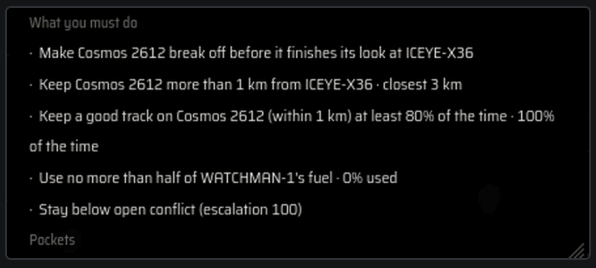
DATABASE F5

With something selected, F5 opens its page. Search on the left (names, types, equipment); `◂ BACK` and the path along the top; every underlined name is a link.

A **craft page** gives its track, class, identity, side, when last seen, its uncertainty and its condition. Your own ships show fuel left. A **class page** gives size and mass, then what the physics makes of the equipment: Δv when full (with the sum written out), thrust, acceleration, burn time; what each sensor can see and how far; what it transmits; how easy it is to see; and how real each number is.

A Red class only appears once you've classified one of its ships. The radar sites each have a page with their limits and sources.

CAMPAIGN F1



`What you must do` : each goal with + done, × failed or · open, and its progress: `closest 3 km`, `100 % of the time`, `0 % used`. The title says the day, `day 1 of 2`.

Below, the **kill chain** for the selected Red track: Heard, Located and identified, Following, Ready to decide, Acted, Out of the fight.

THE SUMS

Click `SNR HERE ⓘ` in the SIGINT station and the **Inspector** opens: each term of the radio link budget, from transmit power through antenna gains, free-space loss and receiver noise to integration over the burst, with its value and the standard behind it, then the total and the chance of hearing it. It follows the live geometry while open.

With doctrinal terms on, hover cards show formulas too.

SOLAR SYSTEM

`Stations › Solar system` : a live chart of the Sun and planets, from JPL's positions, with each one's distance and the cheapest trip there.

HOW COSMOS 2612 BEHAVES

Cosmos 2612 follows simple rules, and knowing them is most of the game. It wants three good looks at X36. It doesn't want company.

ITS PLAN

Three stops, after the real one's reported moves. Each hop takes about 3 hours and two burns (go, then stop):

STOP	WHERE	HOLDS FOR
1	2 km behind X36	6 hours
2	2 km ahead of X36	6 hours
3	0.8 km behind X36	8 hours

Left alone, it finishes in about 29 hours, burns away and you lose. Stop 3 is inside X36's 1 km keep-out, which also costs you points.

WHEN YOU CROWD IT

It keeps **5 km** clear round itself. If WATCHMAN-1 comes within 5 km while it is parked **and it can see you**, it gives way: it moves to a new spot **8 km off, on the far side of X36**, taking about 2 hours, and holds there 4 hours.

It gives way **twice**. The third time you crowd it, it gives up: a 4 m/s burn forward, and it climbs away and drifts off. Once it is 200 km from X36, it has left. (In the tutorial it gives way once and leaves the second time.)

WHEN IT IS ALONE

If it sees none of ours within **50 km** when a stop ends, it moves in: its next spot is at **half the distance**, never nearer than 0.5 km. Stay near and it keeps its distance.

TENSION

Escalation only rises; nothing in CLOSE PASS brings it down.



At 25 it gives way one time fewer, and stops moving in when alone. **At 50** it leaves the next time you crowd it. Pushing hard wins sooner, but costs score and risks war. Warnings appear in the log at 50, 75 and 90.

ITS RADIO

Reports every 3 to 6 hours and after each move, 2 minutes long, on 2,250 MHz, only while its station near Moscow is above the horizon. Quiet for 6 hours after it sees one of ours burn, hears us, or finds us within 10 km.

ITS EYES

It sees WATCHMAN-1 only through its own small infrared telescope. That is why `SEES US` matters: crowding a ship that can't see you does nothing.

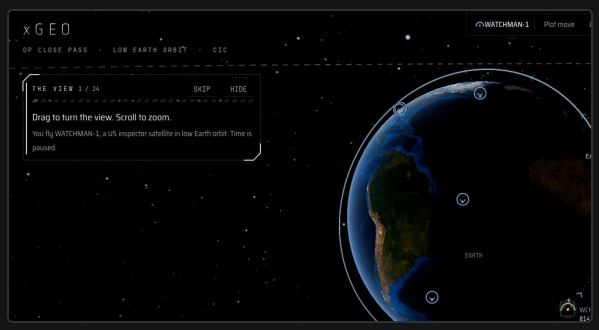


IN THE LOG

"TRK 4423 doesn't want our inspector that close: it's moving to the far side of X36."

24 STEPS

A new game starts with the tutorial card at the top left. Each step names one thing to do and ends when you've done it. There is no Next button: you learn by doing.

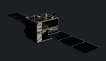


`SKIP` moves on. `HIDE` tucks the card away and leaves a small `TUTORIAL · SHOW` key at the top. Time starts paused. When a wait ends at a high time rate, time stops so you can read the card. Over the SIGINT station the card shrinks to one line.

In the tutorial Cosmos gives way once and leaves the second time it is crowded, so a full run takes less time.

#	STEP	WHAT YOU DO
1	Welcome, operator	Drag to turn the view; scroll to zoom
2	Look anywhere	Right-drag to slide the view
3	The satellite	Click X36 in the Overview
4	Centre on it	Double-click X36
5	The inspector	Click TRK 4423
6	Who's who	Click the Ours tab
7	The database	Stations › Database (F5)
8	Your orders	Esc, then Stations › Campaign (F1)
9	Map layers	Esc, then View › Map layers (L)
10	Your words	Esc, then View › the other Terms
11	Let time run	Speed up to 60× (])
12	The SIGINT station	Stations › SIGINT station (F2)
13	Hear it	Click its newest report when it appears
14	Identify	Press IDENTIFY SIGNAL
15	The link budget	Click the SNR number
16	What you've heard	Esc twice, then Stations › Signals
17	Plot a move	Esc; select TRK 4423; Plot move
18	Go	Pick a move and press Burn
19	Look closer	Hover over TRK 4423
20	The telescope	Look (I); Esc closes it
21	Close in	On arrival, Match orbit within 5 km
22	Every option	Right-click TRK 4423
23	Stay on it	Plot a move to it again; Match on arrival
24	Seen off	Let time run to the debrief

PART 1: GET YOUR BEARINGS



YOUR ORDERS

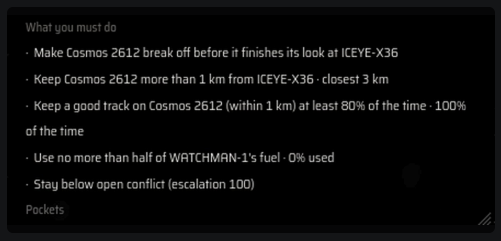
Know where everyone is, and hear it, before you move. Once you're close it stops talking.



1 MEET THE PLAYERS

CLICK X36, THEN TRK 4423, IN THE OVERVIEW

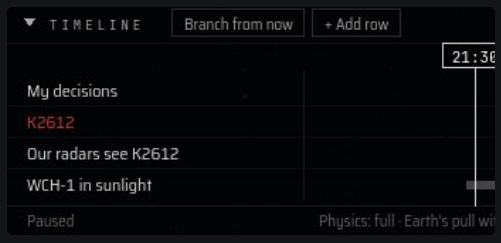
X36 is 20 km ahead of you; TRK 4423 is Cosmos 2612, 15 km away, parked behind X36. Double-click each to see where it is. The Selected item shows TRK 4423's telescope already sees you.



2 READ YOUR GOALS

F1 CAMPAIGN · ESC CLOSES IT

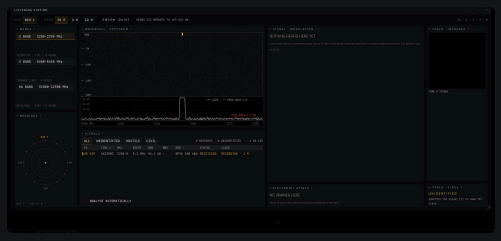
Five goals. The two that lose the game: let it finish its look, or reach escalation 100. The others are points: keep it off X36, keep a good track, save fuel.



3 READ THE TIMELINE

THE TIMELINE, BOTTOM

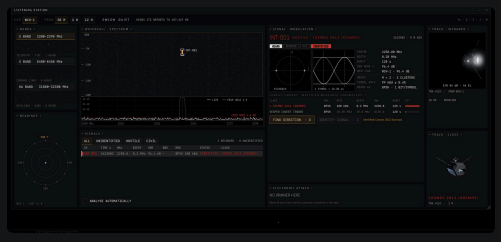
Cosmos's next closest pass to X36, the radar passes, and when you're in sunlight. Add a row for `WCH-1 within 5 km of TRK 4423`: you'll want it later.



4 LISTEN

F2 · S BAND ·] TO 60× OR 600×

Its first report comes within about three hours, while it's over Russia. Watch the waterfall: a burst appears as a bright stripe at 2,250 MHz and the spectrum jumps. When it ends it's logged as INT-001.



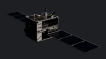
5 IDENTIFY IT

CLICK INT-001 · D FIND DIRECTION · I IDENTIFY

Count the four dots: BPSK. The library matches Cosmos 2612. Now you know who is talking, and the bearing tightened your track. Do it within an hour of the burst.

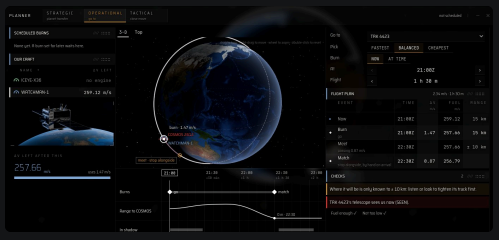
MEANWHILE Cosmos is hopping to its first stop, 2 km behind X36. The log says `TRK 4423 has parked 2 km behind X36` about three hours in. That's your moment.

PART 2: CLOSE IN



YOUR ORDERS

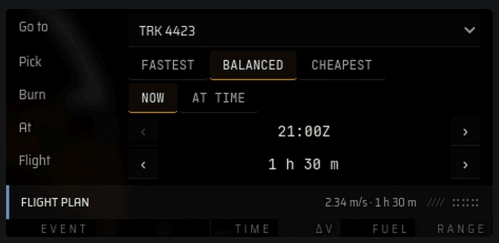
Get alongside it. The planner does the orbital maths; you choose how fast and how much fuel. Arriving isn't stopping.



6 PLOT A MOVE

SELECT TRK 4423 · 1 PLOT MOVE

The planner opens on OPERATIONAL, going to TRK 4423. Read the CHECKS: if the track is only known to ±10 km, a radar pass or a look will tighten it before you go.



7 PICK, THEN BURN

BALANCED OR CHEAPEST · BURN NOW

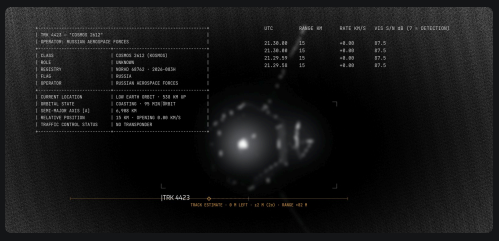
From the start, BALANCED flies in 1 h 30 m for 1.47 m/s and 0.87 m/s to stop. CHEAPEST waits and flies longer for about half that. Either is a tiny slice of your 259 m/s.



8 COAST

] TO 600× · SPACE TO PAUSE NEAR THE END

The move shows on the map as PLANNED MEETING with a countdown. Slow to 60× for the last few minutes.



9 LOOK AT IT

SELECT TRK 4423 · 2 LOOK · ESC CLOSES

In sunlight, close in, the telescope resolves the hull and measures range from its size. In Earth's shadow it sees nothing: wait for the Sun.



10 MATCH ORBIT

3 MATCH ORBIT, ON ARRIVAL

Cancels the last bit of speed so you stay beside it. Stop within 5 km of it, but more than 1 km away.

FUEL About 260 m/s for two days and no refuelling. The fuel goal wants half or less used; full marks at a quarter. Moves of a few metres a second leave plenty.

PART 3: STAY ON IT



TRK 4423 · HOW IT BEHAVES

It keeps 5 km clear round itself. Crowd it while it's parked and it moves away. Leave it alone and it creeps closer to X36.



11 MAKE SURE IT SEES YOU

SELECTED ITEM · SEES US

Crowding only counts if its telescope sees you. A high SEES US means it does. Then you get: `it's moving to the far side of X36`. +5 escalation.



12 FOLLOW IT

1 PLOT MOVE AGAIN · OR TACTICAL FOR SHORT HOPS

It moves to a spot 8 km off on the far side of X36. Plot a move to it once it parks, then Match orbit. For a hop of a few km, TACTICAL pushes of tenths of a m/s are cheapest.



13 MIND THE GAP

STAY MORE THAN 1 KM FROM IT

Each crowding adds 5. Inside 1 km of it, escalation climbs 12 an hour. At 25 it gives way one time fewer; at 50 it leaves the next time you crowd it.



14 SEE IT OFF

CROWD IT A THIRD TIME

It gives up:

`has given up on X36 with our inspector on it, and is leaving.` It climbs away. Once it is 200 km from X36, the operation ends.

IT MOVED CLOSER TO X36?

When none of ours is within 50 km at the end of a stop, it takes its next spot at half the distance. Stay within 50 km and it keeps its distance.

NOTHING HAPPENS?

It gives way only while parked. If it's mid-hop, wait for the log to say it has parked, then close within 5 km.

15 · AFTER THE OPERATION

THE DEBRIEF

When the operation ends the view holds a few seconds on the craft it turned on, under OPERATION COMPLETE, OPERATION FAILED or THE TIME IS UP. Click to skip. Then the debrief.



WHAT IT SHOWS

- **The result** and why: the line that decided it.
- **What you must do**: each goal with a tick, cross or clock, and its progress.
- **Red assets**: TRK 4423, its class, and its fate (still active, or out of the fight).
- **SCORE / 100**, final **ESCALATION**, **DURATION**, debris clouds, then each goal's points, such as `20 of 20`.
- **What happened**: the replay, beside a list of every event about Cosmos (stops, give-ways, burns seen, tracks lost). Click a line to see that moment.
- **Advice** for each goal you missed.

`BACK TO THE TITLE` ends the game and deletes the save.

THE REPLAY

A picture in X36's own frame: X36 at the centre with its 1 km keep-out ring, `→ AHEAD ON ITS ORBIT` and `↑ AWAY FROM THE EARTH`. Your path in blue, Cosmos's in red, the closest moment ringed.

PLAY	The whole mission in 20 seconds; again to pause
CLOSEST	Jump to the closest moment
Slider	Scrub through it
Wheel · drag	Zoom about the pointer · move
Double-click	Fit it all

THE ENDINGS

It leaves	Crowded one time too many, it gave up and left. No weapons, no war.
It finished	All three stops done. Try: get alongside sooner and stay within 50 km.
Open conflict	Escalation reached 100. Try: keep more than 1 km off; crowd only when needed.
Time ran out	Two days, still at work: half marks for the first goal. Try: crowd it earlier.

PLAYING WELL

Things good players learn, most of them the hard way.



LISTEN FIRST

Its reports stop once you're within 10 km or it sees you burn. Identify it while you're still 15 km out.



ALWAYS MATCH

A transfer passes; Match orbit stops. Forget it and you sail by at a metre a second.



BE SEEN WHEN CROWDING

Crowding needs SEES US. Arriving in Earth's shadow may not count until the Sun comes round.



CHEAPEST EARLY

Small moves early cost less than big ones late. Let the orbit do the work.



BRANCH BEFORE BIG MOVES

Branch from now, try it, and go back to Run 1 if it goes wrong.



PICK THE RIGHT SPEED

60× near it, 600× while coasting, 3K× to wait out a stop. Pause to think.



MOVE WHEN IT PARKS

A target mid-hop is a moving goal. Wait for "has parked", then plan.



USE THE SUNLIGHT

Your telescope sees only a sunlit target. The Timeline's sunlight row tells you when.



A TRACK IS A BELIEF

Check FIX before planning. ±10 km means you may arrive somewhere empty.



NOT TOO CLOSE

Between 1 and 5 km is the sweet spot: it gives way, and tension rises only 5.



STAY WITHIN 50 KM

Out of reach, it creeps toward X36 and can enter the keep-out.



MIND THE CLOCK

Three crowdings, each a stop of 4 to 6 hours apart, fit in two days only if you start early.

THREE WAYS TO PLAY IT

THE PATIENT SHADOW

Close in early, stay 2 to 4 km off, crowd it at each stop. Low tension, full custody, a few m/s of fuel. Slow but scores high.

THE HARD PRESS

Crowd it at every chance and sit near 1 km. Escalation passes 25 then 50, and it leaves at the next crowding. Fast, but costs the peace points.

THE GUARD

Park beside X36 instead of Cosmos. It keeps its distance because you're within 50 km, and your telescope holds the track. It never leaves on its own, though: at some point you must go to it.

17 · THE PHYSICS YOU FEEL

WHY ORBITS ARE STRANGE

Everything in xGEO is computed from real physics. You don't need the equations, but these ideas explain what you see.

FALLING ROUND THE EARTH

An orbit is a fall that keeps missing the ground. At 540 km you move at about 7.6 km/s and go round in about 95 minutes. Everything near you is falling the same way, so close in, things barely move relative to each other.

FASTER MEANS SLOWER

Push forward and you climb on the far side of the Earth. A higher orbit takes longer, so you end up **behind** where you started. To catch something ahead of you, you first slow down. The planner handles this; the TACTICAL tab lets you feel it.

ΔV IS FUEL

Fuel is counted as the change in speed it can buy, Δv. The rocket equation sets it from the mass and the exhaust speed: WATCHMAN-1's 75 kg of hydrazine at 2.2 km/s gives about 260 m/s. A move to a craft 15 km away costs 2 or 3 m/s. Going to GEO would cost thousands.

THE EARTH ISN'T ROUND

It bulges at the equator, and the bulge slowly turns every orbit. The Sun and the Moon pull too. xGEO Core includes all three, which is why the planner can solve under full gravity and why paths aren't perfect ellipses.

LIGHT AND SHADOW

Up to a third of each low orbit is in the Earth's shadow. A telescope sees a satellite by the sunlight it reflects, so in shadow it vanishes. Infrared sees heat instead, and heat fades more slowly.

RADIO FOLLOWS THE SQUARE OF DISTANCE

A radio signal spreads out: twice as far, a quarter as strong. Whether you hear a burst depends on the transmitter's power, both antennas, the distance, and your receiver's noise, all in the link budget. Identification needs a clean copy: 15 dB above the noise.

KNOWING IS NOT SEEING

Your track is a filter's best estimate with an honest uncertainty. The tracker is tested over many runs to check that its ring is honest: neither too sure nor too timid. When you lose sight, the ring grows; a radar pass or a look shrinks it.

REAL, PROJECTED, NOTIONAL

Physics and the real satellites' published numbers are **real**. WATCHMAN-1 and the scores are **notional**: invented for play, but everything computed from them follows the physics. Hover cards and the Database say which is which, and the Library gives the sources.

QUESTIONS AND PROBLEMS

THE F KEYS DO SOMETHING ELSE

Click inside the game once. Some browsers keep F1 to F7 until the page has your keyboard. The same commands are on the Stations menu.

MATCH ORBIT IS GREY

Select a contact first, then get within 5,000 km. Hover over it: the card ticks off what's missing.

I CAN'T HEAR ANYTHING

Pick S BAND. Reports come every 3 to 6 hours, only while Cosmos is over Russia, and not at all while you're within 10 km or for 6 hours after it sees you burn. Run the clock at 600× and watch the waterfall.

IDENTIFY SIGNAL WON'T WORK

Wait until the burst has ended. It needs 15 dB: if it's weaker, wait for the next report. And it must be less than an hour old.

THE TELESCOPE SEES NOTHING

The target is in Earth's shadow, or the Earth is behind it. Wait for the sunlight row on the Timeline.

I CROWDED IT AND NOTHING HAPPENED

It must be parked, you within 5 km, and it must see you (SEES US). Mid-hop it ignores you.

THE PLANNER SAYS IT CAN'T

"No way there you can fly" means nothing in the next day is affordable or every path dips too low. Try another destination, or wait.

THE GAME IS SLOW

Close windows you don't need, turn off Ship shadows, use a smaller browser window. At high time rates the clock runs as fast as your computer allows.

WHERE'S MY SAVE?

CONTINUE on the title screen. It's in your browser's storage for this site; clearing site data removes it. Finishing a game deletes it.

SOME WORDS MENTION JUPITER

xGEO first flew at Jupiter, and a few labels still say so. They'll be fixed.

FOUND A BUG?

Use **Report a bug** on the xGEO site. Say what you did, what you expected and what happened, with the mission time.

ESC, IN ORDER

Closes an open menu; cancels an orbit drag; leaves a close-up; closes the telescope; closes the newest window; and with nothing open, saves and goes to the title screen.

CHEAT SHEET

ON ONE PAGE

Print this page and keep it beside the keyboard.

A WINNING ORDER

1. Click X36, then TRK 4423, in the Overview
2. F1 read your goals
3. F2 S BAND ·] to 600×
4. Click its report · D · I
5. Wait for "has parked"
6. Select TRK 4423 · 1 Plot move
7. BALANCED or CHEAPEST · Burn now
8. 2 Look at it in sunlight
9. 3 Match orbit 1–5 km from it
10. It gives way: plot and match again
11. Third crowding: it leaves

WIN AND LOSE

Win	It leaves before it finishes its look
Lose	It finishes three stops · escalation 100
Score	40 leaves · 20 keep-out · 20 track · 10 fuel · 10 peace

TENSION

+5	Each crowding
+12/h	You inside 1 km of it
+3	It parks inside 1 km of X36
25 · 50	Cautious · leaves at next crowding

KEYS

Space	Pause / resume
[ ]	Slower / faster
1 2 3	Plot move · Look · Match orbit
F1	Campaign
F2	SIGINT station
F3	Planner
F4	Overview
F5	Database
F7	Heat forecast
L	Map layers
I V	Telescope · visible / infrared
F	Close-up on the selection
Tab	Next object
W A S D	Turn the view
Q E	Zoom
Esc	Back / close / menu

IN THE SIGINT STATION

↑ ↓	Pick a recording
D	Find direction
I	Identify signal (needs 15 dB)
B	Next band

MOUSE

Drag	Turn · right/middle-drag slide · wheel zoom
Double-click	Focus on it
Right-click	Its orders
Drag your orbit	Plan a burn by hand

WORDS YOU'LL SEE

GLOSSARY

View › Terms switches the game between plain words and military ones. Both are here.

2σ (two sigma) Twice the usual error: a size that holds the truth most of the time. Track rings and FIX are 2σ.

Bearing The direction a signal came from, seen from one ear.

Classify · identify Learn a contact's type from its signal; name the ship itself from a close look.

COP Common operating picture: the shared view of the operation.

Custody Keeping a good track: sure, and right, to within 1 km.

DF Direction finding. Bearings from two places cross at the sender.

Earth's shadow The part of each orbit the Sun can't reach: up to a third of a low orbit.

ES · EW Electromagnetic support (listening) · electromagnetic warfare.

F2T2EA Find, fix, track, target, engage, assess: the kill chain.

Keep-out The space a craft keeps clear round itself. X36's: 1 km. Cosmos's: 5 km.

Apogee · perigee The highest and lowest points of an orbit.

BPSK · QPSK Ways of sending bits by shifting a radio wave's phase. Count the dots to tell them apart.

Contact · track Something you know is there, and your estimate of where it is.

Crowding Parking within 5 km of Cosmos while it is parked and sees you.

dB Decibels: a ratio on a log scale. +3 dB is double, +10 dB is ten times.

Δv (delta-v) The change in speed you can still make: your fuel, in m/s.

EMCON Emission control. QUIET: you only listen. LOUD: you send.

Escalation How close the crisis is to open conflict. 100 ends it.

In-track · radial · cross-track Along the orbit · up and down · sideways. R, I and C.

Lagrange points Five places where the Earth's and Moon's pulls balance.

WORDS YOU'LL SEE

GLOSSARY, CONTINUED

Link budget The sum of gains and losses from a transmitter to a receiver. Gives the SNR.

NORAD number A satellite's number in the public catalogue. Cosmos 2612 is 68762.

SAR Synthetic aperture radar: how X36 images the ground through cloud and at night.

SNR · S/N Signal-to-noise ratio: how far a signal stands above the hiss.

Sun-synchronous A near-polar orbit that crosses each place at the same local time.

Two-body motion Orbits with only one pull. A quick preview; the real burn uses full gravity.

Match orbit The burn that cancels your speed relative to a target, so you stay beside it.

RPO Rendezvous and proximity operations: one craft working close to another.

SIGINT Signals intelligence: listening to other people's radio.

Spectrum · waterfall Signal strength by frequency, now; and the same over time, scrolling down.

TT&C Telemetry, tracking and command: a satellite's radio link with home.

UTC · Z The clock the world's satellites run on. 21:00Z is 21:00 UTC.

LEARN MORE

xGEO Core, the engine under the game, is open source. Its Research Edition describes every model, the sources it rests on and the checks that hold it to them. Both are on the xGEO site.

xGEO Live Operator's Handbook · Alpha 0.5, build 11 · November 2026.

© 2026 Kevin Williamson. All rights reserved. xGEO is an independent educational project. ICEYE-X36 and Cosmos 2612 are real satellites, flown from their public orbits and specifications, and the ground radars are real sites. WATCHMAN-1 is invented, and what happens in the game is fiction. Their operators have not reviewed or endorsed this game.